

Group 2 - Project 2

Customer, Product & Competitor Knowledge Graph + GraphRAG

Group Members	Business Areas	Technology Themes
Ajok Aditya S Varun Trailokesh Mohanty	Customer Analytics; Consumer Insights; Segmentation; Personalization; Market & Competitive Intelligence; Product and competitor relationship discovery	GraphRAG, NLP NER, relationship extraction, entity resolution, graph database, Cypher generation, graph embeddings, natural-language Q&A over graph + documents

Project Objective

Build a knowledge graph that connects customers, segments, products, brands, competitors, campaigns, purchase drivers, complaints, and market signals. The project should demonstrate how graph-based reasoning can reveal non-obvious relationships that plain document search or vector similarity often misses, such as which competitor is gaining with which segment and why.

High-Level Work Areas

1. Define graph schema: CustomerSegment, Product, Brand, Competitor, Campaign, NeedState, ReviewTheme, Channel, Market, Study, and Insight.
2. Extract entities and relationships from reviews, surveys, reports, and product catalogs.
3. Resolve duplicate entities and normalize brand/product names.
4. Load graph into Neo4j or equivalent graph DB.
5. Build Cypher queries for common business questions.
6. Combine graph traversal with document retrieval for GraphRAG answers.
7. Compare vector-only RAG versus GraphRAG on relationship-heavy questions.
8. Build explainability view showing graph path and supporting evidence.

Input Data / Assets

Customer feedback samples, product catalogs, competitor profiles, campaign metadata, e-commerce reviews, social listening themes, segmentation summaries, market research snippets

Evaluation Metrics

Entity extraction F1, relationship extraction precision, entity resolution accuracy, graph query correctness, path explainability, GraphRAG answer faithfulness, analyst usefulness

Progressive Demo Sequence

The sequence is designed to progressively expose the limitations of a base approach, then add capabilities that improve relationship discovery and evidence-backed reasoning.

1. Start with a question: 'Which competitor is gaining among value-conscious parents, and what product attributes are driving the shift?'
2. Show vector-only RAG retrieving several relevant documents but failing to connect segment -> competitor -> product attribute -> channel.
3. Show manual Cypher/graph query on a small prebuilt graph; highlight that graph traversal answers relationship questions but lacks natural-language explanation and document evidence.
4. Add NER extraction from reviews/reports for entities such as segment, brand, product, attribute, complaint, and channel. Show extraction errors and why human validation/schema design matters.
5. Add relationship extraction: 'Segment prefers Product', 'Competitor wins on Attribute', 'Campaign targets Segment'. Show graph population.
6. Add entity resolution; demonstrate that 'Brand-A', 'Brand A', and SKU aliases need normalization.
7. Ask GraphRAG: 'Show the evidence chain for why Competitor X is gaining in Segment Y.' Display graph path plus cited document snippets.
8. Compare vector RAG versus GraphRAG on: 'Which unmet need connects complaints about price, packaging, and competitor switching?' Show graph advantage.
9. Add graph + vector hybrid retrieval for questions needing both structured relationships and unstructured explanation.
10. Final limitations to highlight: graph quality depends on extraction quality, relationship direction can be ambiguous, temporal changes are hard, and graphs can become stale.
11. Future scope: temporal knowledge graph, human-in-the-loop graph curation, active learning for extraction, graph embeddings, causal analysis, and automated graph evaluation.

Compact Project Snapshot

Use Case	Customer, Product & Competitor Knowledge Graph + GraphRAG
Core Value	Reveal relationship-heavy insights that vector similarity alone may miss.
Key Question	Which competitor is gaining in a segment, through which attributes/channels, and why?
Differentiator	Graph paths + document evidence + natural-language explanation.
Main Risks	Extraction quality, entity resolution, ambiguous relationship direction, stale graphs.
Future Scope	Temporal KG, human-in-the-loop curation, active learning, graph embeddings, causal analysis.